

DATA STRUCTURES AND ALGORITHMS ASSIGNMENT

For practice only – solve at home

Multiple Choice Questions

1. Let A be a two-dimensional array declared as follows:

A: array [1...10] [1...15] of character;

Assuming that each character takes one memory locations the array is stored in row-major order and the first element of the array is stored in location 100, what is the address of the element $A[i][j]$?

- (a) $15i + j + 84$
 - (b) $15j + i + 84$
 - (c) $10i + j + 89$
 - (d) $10j + i + 89$
2. Given a pointer to the first node of a singly linked list, which of the following operation can be implemented in $O(1)$ time?
- (I) Insertion at the front of the linked list
 - (II) Insertion at the end of the linked list
 - (III) Deletion of the front node of the linked list
 - (IV) Deletion of the last node of the linked list
- (a) I and II
 - (b) I and III
 - (c) I, II and III
 - (d) I, II and IV
3. Linked Lists are not suitable for:
- (a) Binary Search
 - (b) Polynomial Manipulation
 - (c) Insertion Sort
 - (d) Radix Sort
4. In the worst case, the number of comparisons needed to search for a key in a singly linked list of length n is:
- (a) $\log_2 n$
 - (b) $n/2$
 - (c) $\log_2 n - 1$
 - (d) n
5. The time complexity of inserting an element at the beginning of a singly-circular linked list having only one external pointer pointing to the head is:
- (a) $O(1)$
 - (b) $O(\log_2 n)$
 - (c) $O(n)$
 - (d) $O(n^2)$
6. Deleting a node at the end of a double linked list, where head & tail both are known would take time:
- (a) $O(1)$
 - (b) $O(n)$
 - (c) $O(\log n)$
 - (d) $O(n \log n)$

7. Which of the following statements is correct for a circular singly linked list with only a start pointer?

- (a) Both insertion and deletion at the front end take $O(1)$ time
- (b) Only insertion at the front end takes $O(1)$ time
- (c) Only deletion from the front end takes $O(1)$ time
- (d) No insertion or deletion operation at either end is possible in $O(1)$ time

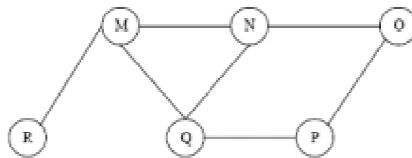
8. What is the output of following function for start pointing to first node of the following linked list:

1->2->3->4->5->6?

```
void fun(struct node* start)
{
    if(start == NULL)
        return;
    printf("%d ", start->data);
    if(start->next != NULL )
        fun(start->next->next);
    printf("%d ", start->data);
}
```

- (a) 1 4 6 6 4 1
- (b) 1 3 5 1 3 5
- (c) 1 2 3 5
- (d) 1 3 5 5 3 1

9. The Breadth First Search algorithm has been implemented using the queue data structure. One possible order of visiting the nodes of the following graph is



- (a) MNOPQR
- (b) NQMPOR
- (c) QMNPRO
- (d) QMNPOR

10. What will be the post order traversal of the given tree?



- (a) 1, 2, 3, 4, 5, 6
- (b) 5, 3, 4, 6, 2, 1
- (c) 4, 3, 6, 5, 2, 1
- (d) 3, 4, 6, 5, 2, 1

11. Which of the following tree can always be stored with optimum space complexity, using a 1D array?

- (a) Full Binary Tree
- (b) Almost complete Binary Tree
- (c) Binary Search Tree

12. The number of edges in a complete undirected graph with n vertices is

- (a) $n(n-1)$
- (b) $n(n-1)/2$
- (c) n^2
- (d) $2n-1$

13. Which of the following statement(/s) is (/are) correct?

Statement 1: Each B-Tree of order 2 will always be a Binary Search Tree

Statement 2: Every graph will always be a tree.

Statement 3: Every tree will be a graph, but every graph will not always be a tree

Statement 4: Every graph will be a tree, but every tree will not always be a graph.

- (a) Both Statement 2 and Statement 4
- (b) Only Statement 1
- (c) Both Statement 1 and Statement 3
- (d) Only Statement 3

14. In a full binary tree if there are L leaves, then total number of nodes N is

- (a) $N = 2 * L$
- (b) $N = L + 1$
- (c) $N = L - 1$
- (d) $N = 2 * L - 1$

15. In array representation of a binary tree if index number of a node is 7 then index number of its right child is

- (a) 12
- (b) 13
- (c) 15
- (d) 14

16. Let G be an undirected graph. Consider a depth-first traversal of G , and let T be the resulting depth-first search tree. Let u be a vertex in G and let v be the first new (unvisited) vertex visited after visiting u in the traversal. Which of the following statements is always true?

- (a) $\{u, v\}$ must be an edge in G , and u is a descendant of v in T
- (b) $\{u, v\}$ must be an edge in G , and v is a descendant of u in T
- (c) If $\{u, v\}$ is not an edge in G then u is a leaf in T
- (d) If $\{u, v\}$ is not an edge in G then u and v must have the same parent in T

17. The equivalent Postfix notation for $((A + B) * C - (D - E) ^ (F + G))$ is:

- (a) $AB+C*DE-FG+^--$
- (b) $^--*+ABC-DE+FG$
- (c) $(AB)+CDE*--FG+^$
- (d) $AB*+DE--FG+^$

18. What is the type of the algorithm used in solving the 8 Queens problem?

- (a) Divide and conquer
- (b) Backtracking
- (c) Greedy
- (d) None of the above

19. What is the data structure used to implement the BFS traversal of a graph?

- (a) Stack
- (b) Queue

- (c) Binary Tree
- (d) B-tree

20. Which of the following operations on a queue implemented by an array in a naive way has time complexity $O(n)$, where 'n' is the number of elements in the queue?

- (a) Enqueue
- (b) Dequeue
- (c) Push
- (d) isEmpty

21. When using a stack to convert an infix expression into its postfix equivalent, which of the following is true?

- (a) The stack may contain both opening and closing brackets
- (b) A plus operator can be pushed on top of a binary minus operator
- (c) A division operator can be pushed on top of a plus operator
- (d) An open bracket can be popped off the stack when a division operator is processed

22. Recursion uses the ___ data structure as it follows ___ ordering.

- (a) Queue, LIFO
- (b) Queue, FIFO
- (c) Stack, LIFO
- (d) Stack, FIFO

23. What would be the Postfix notation for the given equation?

$$a + (b * c (d / e^f) * g) * h$$

- (a) $abcde f^/*g*h*+$
- (b) $ab*cdef/^*g-h+$
- (c) $abc*de^fg/*-h+$
- (d) $abcd*^ed/g*-h*+$

24. A stack S has the entries a,b,c with 'a' on the top. Another stack T is empty. An entry popped out of the Stack S can be printed immediately or pushed in to the stack T, finally popped out of the Stack T and printed. Then which sequence can never be printed?

- (a) a b c
- (b) b a c
- (c) b c a
- (d) c a b

25. Which of the following is true regarding a Max Heap?

- (a) Maximum element occurs at the root
- (b) Minimum element occurs as the last child element
- (c) BuildMaxHeap has time complexity $O(n \log n)$
- (d) HeapSort can have worst case time complexity as $O(n^2)$

26. When does the worst case complexity happen for Insertion Sort?

- (a) When the array is already sorted
- (b) When the array is sorted in reverse order
- (c) When the array is not sorted
- (d) When the array is sorted till the midpoint and then reverse sorted

27. Interpolation search fails when

- (a) All values are within a small range
- (b) All values are uniformly distributed

- (c) All values are non-uniformly distributed
(d) Never
28. Binary search can be applied to an array that is
- (a) Not sorted
(b) Already sorted
(c) Having random values
(d) Of any ordering
29. If all elements in a heap are identical, then the heap is:
- (a) A min-heap but not a max-heap
(b) A max-heap but not a min-heap
(c) Both a max-heap and a min-heap
(d) Neither a max-heap nor a min-heap
30. If an array has n keys which are uniformly distributed, the number of comparisons required to locate a key x that is already present in the array is:
- (a) 1
(b) $\sqrt{n}$
(c) $n/2$
(d) $\log_n x$
31. Which of the following sort algorithms will run the fastest when applied on an unsorted linked list?
- (a) Optimised Bubble Sort
(b) Cocktail Shaker Sort
(c) Insertion Sort
(d) Selection Sort
32. The time complexity of running binary search on a linked list whose elements are already sorted is:
- (a) $O(1)$
(b) $O(\log n)$
(c) $O(n)$
(d) $O(n \log n)$

Fill in the gaps

- Linked lists are not suitable to implement _____ search.
- The order of the triplet array representation of a 5×10 matrix which has 15 non-zero elements is _____.
- A linked list having n nodes ($n \geq 1$) where no node stores a NULL pointer is a _____ linked list.
- The time complexity of displaying the contents of a singly-linked list in reverse order without using a stack is _____.
- If front is equal to rear in a circular queue which is not empty, the queue has _____ element(s).
- The optimal data structure used to solve the Tower of Hanoi problem is _____.
- Total number of unique binary search trees formed with the nodes 10, 20, 30 is _____.
- For any B-tree of minimum degree $t \geq 2$, every node other than the root can have at most _____ keys.
- The number of leaves in a complete binary tree of height h is _____.
- If a node X is both the leftmost and rightmost node in a binary search tree, then the degree of X is _____.
- If the in-order traversal of a tree is E A C K F H D B G, its pre-order traversal is _____.
- The minimum number of edges in a connected undirected graph, having n nodes is _____.
- Can a binary search tree be called a max-heap (Yes/No) _____.
- The collision resolution technique where the colliding element is placed in the next available empty slot in the hash table is called _____.

15. The number of swaps required by selection sort is (greater/lesser) _____ than bubble sort.
16. The time complexity of building a max-heap is _____.

Subjective Questions

DATA STRUCTURE BASICS

1. Define Big-Oh, Big-Omega and Big-Theta with example. Also draw their asymptotic curves.
2. Which one is asymptotically bigger, n or $\log n$? Draw the graphs required.
3. Given a 2D integer array of dimension 100×200 and also given that the base address of the 2D array is 2020. Assume that the array index of the matrix starts from $[10, 20]$. Find the address of the element with index $[50, 160]$ for column major ordering and row major ordering.
4. Find the Big-Oh for $T(n) = 48n^{100} + 2^{n+2} + 3n^2 + 100$.
5. What is the difference between an abstract data type (ADT) and a data structure?

ARRAY APPLICATIONS

6. What is a sparse matrix?

Given the following 4×5 matrix:

```
0 0 6 0 0
0 0 0 0 0
4 0 2 0 0
0 3 0 7 0
```

Compute its triplet array equivalent.

Is it beneficial to store this matrix as a triplet array? Justify your answer.

7. Propose an algorithm to remove duplicates from an ordered array, without using any second array. For example, if the original content of the array is: 1, 1, 3, 4, 8, 8, 10, 10, 25, 28, 30, 30, 30; then the final content of the array should be 1,3,4,8,10,25,28,30. What is the time complexity of the proposed algorithm?
8. If the number of nonzero elements in the triple representation of a sparse matrix of $m \times n$ dimension is K , then what is the time complexity to read an element from the index $[i][j]$ of the sparse matrix. Assume that the sparse matrix is stored in triple format.
9. How can you reverse an array without using an extra array? Provide the algorithm and analyze its time and space complexity.
10. Evaluate the result of the given infix expression using proper data structure:
 $2 * (5 * (3 + 6)) / 15 - 2$.
Show every step clearly.
11. Stepwise show how the following infix expression is converted to its corresponding postfix expression.
 $((5 + 7) - (4 * 3)) + (6 / 3) / 2 - 9$
Next, Use a suitable data structure to evaluate the generated postfix expression.
12. Write the pseudo-code of insertion and deletion of an element from an input-restricted double-ended queue. The deque is implemented using circular array.

13. Write an algorithm to merge eight sorted queues into one single queue in such a way that duplicates are removed and the resultant queue is sorted also. Comment on the time complexity of your algorithm. More credits will be given to more efficient algorithms.
14. Write an algorithm for inserting into an input restricted deque. Show the working of this algorithm by the input sequence: 1, 2, 3, 4, 5, 6.
15. Using a suitable data structure, write a C function to determine whether a sequence of parentheses is balanced or not.
16. There are 'n' integers in a circular queue. We search for every 'd'-th element starting from the queue beginning and mark it invalid. So after the first round, the 'd'-th element is marked invalid and there are only (n-1) valid items remaining in the queue. Starting from the first invalid element, this process continues, i.e., every 'd'-th element is searched and marked invalid. Invalid items are no longer taken into account and skipped. That means if you come across an invalid item when you try to find the 'd'-th element, you try the 'd+1'-th element, if that is valid. If invalid, you try 'd+2'-th element. This continues till you find the first valid item or terminate if no valid items could be found. Find after 'x' rounds, the element that will be marked invalid and return the element.
17. There are n elements in a stack, stored in a sorted way. Move these elements to another stack so that the final stack is always sorted during the entire movement. You may additionally use another stack but the sorted property has to be maintained there all the time. An iterative version will fetch you additional credits.
18. Suppose you want to implement a queue and a stack using an unrestricted deque. Describe the deque methods that need to be reused to implement standard stack and queue operations.
19. Implement two stacks on the same array, managing space most efficiently.
20. Give an example of an output sequence that cannot be generated from an input sequence via a series of push and pop operations on an initially empty stack.
21. The span of a stock's price on a certain day, d, is the maximum number of consecutive days (up to the current day) the price of the stock has been less than or equal to its price on d.

An example is give below:

Day #	1	2	3	4	5	6
Price	9	3	5	2	7	12
Span	1	1	2	1	4	6

A naive comparison based algorithm will run in $O(n^2)$ time. You have to design an algorithm using stack so that the algorithm can be run in $O(n)$ time.

RECURSION

22. Discuss the key features of tail recursion. Explain with suitable example how it is different from a normal recursive function.
23. Write a recursive function to determine whether a given string is a palindrome. Draw the recursion tree when the string "ABCBA" is passed to the function.

24. Write a small C function (no need for main, header files etc.) to print the moves of the well-known Tower of Hanoi problem using recursion. Assume that the smallest disc is numbered as 'disc 1' and the largest is 'disc n'. Each of the three pegs is represented by a char (datatype): source ('S'), destination ('D') and temporary ('T'). Given below are some examples of the moves that you will print:

Move the disc # 12 from T to D

Move the disc # 4 from S to T.

Now just write the output of your function if it is called with $n = 4$, where n is the number of discs.

25. What will the following function return if it is called from the main as `fun(5, 0, 1)`? Can you identify what the following function is calculating? It is something well-known.

```
int fun(int n, int a, int b)
{
    if (n == 0)
        return a;
    if (n == 1)
        return b;
    return fun(n - 1, b, a + b);
}
```

26. Explain what will be the output when `fibonacci(5)` is called:

```
void fibo(int n) {
    if (n==0) or (n==1) return;
    printf("Inside fibo(%d)\n",n);
    fibo(n-1);
    fibo(n-2);
    return;
}
```

LINKED LIST

27. Write a function in C/ pseudo-code to delete all the nodes that contain the certain value v , from a singly linked list in a single pass. If there are multiple occurrences, delete all occurrences. Discuss the worst case time complexity of the algorithm.
28. Write the pseudo-code of inserting an element after a given value in a doubly linked list. Your code should work for any position of the given value. Show proper error messages, if any.
29. Given a singly-linked list with an odd number of elements, develop a pseudo-code or C-code to split it into two nearly equal sub-lists — one for the front half, and the other for the back half. The extra element should go in the front list.
30. Write the pseudo code to insert an element into a singly linked list at a given position. Take appropriate measure when the list is empty or if the given position does not exist in the list.
31. Write an algorithm/C-like pseudo code to delete the last element from a circular singly linked list. What is the time complexity of this operation?
32. Write a function which will take any number n as its argument. The function will break this number into its individual digits and then store every single digit in a separate node thereby forming a linked list. The function must return the head node address of the created linked list at

the end. (For example, if the number is 13579, then there will be 5 nodes in the list containing nodes with values 1, 3, 5, 7, 9).

33. How can you represent the polynomial $5x^5 + 4x^2 - 25x + 10$ with array(s)? Additionally, if a singly linked list is used to represent this polynomial instead, would the arithmetic operations like addition/subtraction become more efficient? Provide your analysis.
34. How do you merge two sorted linked lists into a single sorted linked list? Provide an algorithm and analyze its time complexity.
35. Suggest a pseudo code / C function to find the middle element of a single linked list in a single pass. (Please note that if the list has even number of elements (n) then middle element will be the first middle element ($\lfloor n/2 \rfloor$).
36. How is a circular linked list with a head pointer different from a circular linked list with a tail pointer? Which of the two would be better suited for implementing a queue and why, provided the list is allowed to have only one external pointer?
37. Implement a Double Ended Queue using a doubly linked list.
38. Write the pseudo code to determine whether a singly linked list of integers is a palindrome.

TREE & GRAPH

39. Define Binary Search Tree.

Consider a Binary Search Tree (BST) consisting of distinct integers lying in the range between 1 and 1000. In order to search (or probe) the number 363 in the given BST, which of the following search (or probe) sequences is correct one and which sequences are not correct. Justify your answer for each of these search (or probe) sequences.

- (i) 2, 252, 401, 398, 330, 344, 397, 363
 - (ii) 924, 220, 911, 244, 898, 258, 362, 363
 - (iii) 925, 202, 911, 240, 912, 245, 363
 - (iv) 2, 399, 387, 219, 266, 382, 381, 278, 363
 - (v) 935, 278, 347, 621, 299, 392, 358, 363
40. The square of a directed graph $G=(V,E)$ is the graph $G^2=(V,E^2)$ such that (u,w) in E^2 if and only if for some v in V , both (u,v) in E and (v,w) in E . In other words, G^2 contains an edge between u and w whenever G contains a path with exactly two edges between u and w . Describe the outline of an efficient algorithm for computing G^2 from G for the adjacency-matrix representation of G . Analyze the running times of your algorithm
 41. What are the similarities and differences between an AVL tree and a B-Tree of order 2.
 42. Consider a binary tree with n (i.e., $n \geq 0$) nodes, which has been constructed using linked representation, where the structure of each tree node has three components: data, pointer to left subtree and pointer to right subtree. Suggest an algorithm (/ pseudo code / C-code) of a function (or procedure) which will determine whether the given binary tree is a complete or an almost complete binary tree or not.

[N.B. Assume that the function has access to the root node of the given binary tree.]

43. Given an adjacency-list representation of a multi-graph $G = (V, E)$, describe the outline of an algorithm to compute the following representations of the "equivalent" undirected graph $G' = (V, E')$, where E' consists of the edges in E with all multiple edges between two vertices replaced by a single edge and with all self-loops removed :

- (i) Adjacency-list of undirected graph G' .
 - (ii) Adjacency matrix representation of undirected graph G' .
- Analyze the running times of the above algorithms.

44. Consider a binary tree with n (i.e., $n \geq 0$) nodes, which has been constructed using linked representation, where the structure of each tree node has three components: data, pointer to left subtree and pointer to right subtree. Suggest an algorithm (/ pseudo code / C-code) of a function (or procedure) to determine that whether the given binary tree is a Binary Search Tree (BST) or not.

[N.B. Assume that the function has access to the root node of the given binary tree.]
What is the best case time complexity of the above algorithm?

45. Consider a binary tree with n (i.e., $n \geq 0$) nodes, which has been constructed using linked representation, where structure of each tree node has three components: data, pointer to left subtree and pointer to right subtree. Write a recursive algorithm that will display all the keys of a given Binary search tree in sorted (ascending) order.

46. Given an undirected graph $G=(V,E)$, let $d(u, v)$ be the minimum number of edges on a path between u and v . Let the diameter of G be the maximum $d(u, v)$ over all such pairs (u, v) in G . Explain how you may find the diameter of a graph using BFS.

47. Show each and every steps to construct a binary search tree (BST), for which the in-order and post-order traversal sequences are as follows:

In-order : 20 30 50 75 80 100 180 190 195 200 250 295
Post-order: 30 20 75 80 50 195 190 200 180 295 250 100

Now delete the following data (one at a time) 100 and 50 from the above constructed BST.

48. Show that, the height of a binary tree that contains n elements ($n > 0$), is at most $n-1$ and at least $\lceil \log_2 n \rceil$.

49. Describe the outline of an algorithm that uses DFS and back edges to detect cycles in an undirected graph. Analyze the running time of your algorithm. Is it possible to run this in $O(V)$ time rather than $O(V+E)$?

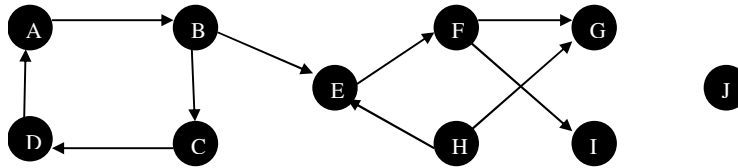
50. Consider a binary tree with n (i.e., $n \geq 0$) nodes, which has been constructed using linked representation, where structure of each tree node has three components: data, pointer to left subtree and pointer to right subtree respectively.

Suggest a recursive algorithm (/ pseudo code / C-code) of a function (or procedure) which will determine the height of the given binary tree.
[N.B. Assume that the function has access to the root node of the given binary tree.]

51. Show each step of insertions, to construct a B-tree of order 3, where the following keys are inserted one at a time, as per the given sequence:

10, 20, 30, 40, 70, 65, 66.
After construction delete 40 from the B-tree.

52. Perform a DFS traversal on the following graph starting from vertex H, i.e., $d(H) = 1$. Just write the discovery time $d(u)$ and finishing times $f(u)$ for each vertex u . You need not show any color change or intermediate versions of the graph. Also classify the edges into the four different types – tree, back, forward and cross edges.



53. Briefly explain the difference between a full binary tree and a complete binary tree with suitable examples.

54. Consider a binary search tree with n (i.e., $n \geq 0$) nodes, which has been constructed using linked representation of tree, where the structure of each tree node has three components: key value/ data, pointer to left subtree and pointer to right subtree.

Suggest an algorithm (/ pseudo code/ C-code) of a function (or procedure) which will delete a given key value from the binary search tree, provided it is found in the tree

55. Describe the outline of an algorithm for topological sort on a DAG which operates by repeatedly finding a vertex of zero in-degree and removing all its outgoing edges. Assume an adjacency matrix representation. Analyze the running time of your algorithm. Is your algorithm faster than one using DFS?

56. Define a B-Tree of order m .

57. Show each step of insertions (along with any height balancing steps, if required), to construct an AVL tree using the keys, inserted one at a time, as per the following given order:

45, 80, 70, 40, 30, 65, 35, 25, 27, 15.

Now show each steps of deletion (along with any height balancing steps, if required) from the above constructed AVL tree, where each of the following keys to be deleted (one at a time): 70, 80.

58. You are trapped in a network of tunnels with only one exit. The network is connected in that all points are reachable. You have a box of chalk pieces to mark on the walls. Explain how you can come out of the tunnel network by searching for the exit efficiently. Analyze the running time of your algorithm.

59. Consider a binary tree with n (i.e., $n \geq 0$) nodes, which has been constructed using linked representation of tree where structure of each tree node has three components: data, pointer to left subtree and right subtree respectively.

Suggest an algorithm (/ pseudo code / C-code) of a function (or procedure) which will determine whether the given binary tree is a full binary tree or not.

[N.B. Assume that the function has access to the root node of the given binary tree.]

60. Briefly discuss about in-threaded binary search tree using a suitable example.

61. Given an undirected graph $G = (V, E)$, and a designated source vertex s , we wish to find the closeness centrality of s . This is defined as the average over all other vertices v , the minimum

number of edges between s to v . Explain how closeness centrality can be computed using BFS. Analyze the running time of your algorithm.

62. Write an iterative function (not recursive) to traverse the nodes of a binary search tree in in-order sequence.

SEARCHING, SORTING & HASHING

63. When do you say that a certain sorting algorithm is stable?

The following pseudo-code works on an array of records indexed from 1 to n , where k_j is the key of the record of type integer present at index j . What job does it perform?

```
for i = n - 1 downto 1
  for j = 1 to i
    if  $k_j > k_{j+1}$ 
      do exchange
    endif
  endfor
endfor
```

What changes occur (if any) if you change the $\leq$ operator in the above pseudo-code to

- (i) $<$
- (ii) $\geq$
- (iii) $>$

64. What is the greatest drawback of the naïve bubble sort algorithm that we cannot remove even if we use its improved version?

However, if we do a small modification in the algorithm, it gets removed. What is that modification?

What do we call the sorting algorithm after we perform that modification?

65. What is achieved in each pass of a Selection sort algorithm where the elements are stored in an array starting with index 1? Just answer in one line saying "In the k^{th} pass of the Selection Sort algorithm _____ is brought to position _____."

Do an asymptotic time-complexity analysis for finding the number of comparisons done by the Selection sort algorithm.

66. (i) Given below is a typical pseudo-code for insertion sort that works on an integer array A indexed from 1 to n and sorts it in ascending order:

```
ALGO- INSERTION-SORT(A)
INPUT- Array A[1..n]
1. for j ← 2 to n
2.   do key ← A[j] //Insert into sorted A[1...j-1]
3.     I ← j - 1
4.     while (I > 0) and (A[I] > key)
5.       do A[I+1] ← A[I]
6.       I ← I - 1
7.     A[I+1] ← key
```

How many times does each of the lines numbered 1 to 7 execute? As for example, line no. 1 gets executed $(n - 1)$ times. You may additionally assume that the line 4 executes t_j number of times when the index variable of the for loop takes the value j .

- (ii) For what sort of input pattern, the best-case of insertion sort will occur? What will be the value of t_j then for each j ?

For what sort of input pattern, the worst-case of insertion sort will occur? What will be the value of t_j then for each j ?

Do a probabilistic analysis to show that the value of $t_j = (j + 1)/2$ in the average case assuming a uniform distribution for the input.

(iii) Assuming that the cost of executing line no. 1 is C_1 , line no. 2 is C_2 and so on, and using the number of times each line gets executed (what you found out in part (a)), do a worst-case analysis for insertion sort to show that the total time taken is quadratic in n .

67. Can you give a suitable example array with 10 integers on which interpolation search will have the same worst-case time complexity as linear search.

68. What is the asymptotic time complexity of Binary Search in the average case? Do the average case time complexity analysis for Binary Search, in case the element is surely present in an ordered array of size n .

69. Explain briefly with a simple diagram how closed hashing is done. What must hold assumption in such a data structure to ensure that each of the hashing operations remain $O(1)$.

70. We know that the recurrence relation defining the time required for Binary Search is $T(n) = T(n/2) + 1$. Write a similar recurrence relation for the average case of interpolation search.

State the asymptotic time-complexity of interpolation search. Derive it from the recurrence relation that you defined above.

71. Assume that we are talking about sorting in descending order. What are the best-case and worst-case **inputs** for improved bubble sort?

What are the best-case and worst-case asymptotic **time complexities** for Selection sort?

72. Show that the expected number of comparisons required to find an element from an array of n elements using sequential search, if the element occurs twice in the array, is $(n + 1) / 3$.

73. In the following pseudo-code for Heapsort, it is assumed that the array index starts from 1. Can you write the modified pseudo-code for these two procedures in case the array index starts from 0?

```
BUILD-MAX-HEAP( $A$ )
1   $heap-size[A] \leftarrow length[A]$ 
2  for  $i \leftarrow \lfloor length[A]/2 \rfloor$  downto 1
3      do MAX-HEAPIFY( $A, i$ )

HEAPSORT( $A$ )
1  BUILD-MAX-HEAP( $A$ )
2  for  $i \leftarrow length[A]$  downto 2
3      do exchange  $A[1] \leftrightarrow A[i]$ 
4           $heap-size[A] \leftarrow heap-size[A] - 1$ 
5          MAX-HEAPIFY( $A, 1$ )
```

74. If a Heap is stored in an array starting with index = 1,

- i) What will be the index of the left child of a node whose index is 5?
- ii) What will be the index of the right child of a node with index 21?
- iii) What will be the index of the parent of a node, whose index is j ?
- iv) What are the possible number of elements in the Heap if the index of the rightmost parent is 100?

75. Write the non-recursive pseudo-code for Binary-search. State its best-case and average case complexity.

76. Give an example of a situation where using counting sort is better than using any other comparison-sort technique? State the asymptotic time-complexity of counting sort, clearly defining the parameters that you use in your expression.

77. In a closed hashing we are using an array of size 10 whose 0, 1, 2, 5, 6, 9 indexed positions are already filled and the hash function is $h(k) = (k + 10) \% 10$. First draw the array in such condition with some probable values. Now calculate the probability, that the next key which is going to be inserted should **NOT** be placed at the position with index = 3.

78. Implement an algorithm to perform insertion sort on a singly-linked list which does not increase in time complexity when compared with the array equivalent. Hint: Traversing the partially-sorted portion of the list from right to left increases the time complexity. Try modifying the original algorithm so that the traversal can be done from left to right instead.

79. i) What is the expression for Load factor (α) in the context of Hashing by Open Addressing? Also, write what the terms signify that you use in the expression for α .

The average cost for linear hashing in case of –

Insertion is $1/2(1 + 1/(1 - \alpha)^2)$ and deletion is $1/2(1 + 1/(1 - \alpha))$

ii) Which one do you think requires more time – insertion or deletion?

iii) Show that your above answer follows from the above two expressions.

iv) What could be the rationale behind it?

80. Fill up the following table with the appropriate asymptotic complexities:

Type of Sorting	Best Case	Average Case	Worst Case
Naïve Bubble Sort	$O(n^2)$		
Improved Bubble Sort			
Insertion Sort			
Selection Sort			$O(n^2)$
Counting Sort		$O(n + k)$	